

ENZYMES

:Oxidoreductases include .1

- a) Catalases and Mutases
- b)** Oxygenases and Dehydrogenases
- c) Dehydrogenases and Ligases
- d) Aminotransferases and Racemases
- e) Epimerases and Oxidases

:Michaelis constant of enzyme (Km) is .2

- a) The substrate concentration which gives maximal velocity
- b)** The substrate concentration which gives 1/2 the maximal velocity
- c) 1/2 the substrate concentration which gives maximal velocity
- d) The enzyme concentration which gives maximal velocity
- e) The enzyme concentration which gives 1/2 the maximal velocity

:A competitive inhibitor .3

- a)** Increases the Km of the enzyme
- b) Decreases the Vmax of the enzyme
- c) Increases the Vmax of the enzyme
- d) Decreases the Km of the enzyme
- e) Increases both Km and Vmax of the enzyme

:The non-protein part of an enzyme is called .4

- a) Apoenzyme
- b) Holoenzyme
- c) Allosteric Enzyme
- d) Isoenzyme
- e)** Coenzyme

:The major 6 classes of enzymes include .5

- a) Transferases and Oxygenases
- b)** Isomerases and Lyases
- c) Hydrolases and Epimerases
- d) Oxidoreductases and Catalases
- e) Transaminases and Transferases

:Enzymes .6

- a) Increase the energy barrier for a reaction
- b) Are recovered chemically altered after a complete reaction
- c) Increase the equilibrium constant of the reaction
- d)** Activity is subject to regulation
- e) Are exclusively proteins

Coenzymes .7

- a) The non-protein part of enzymes
- b) Needed for enzyme activity
- c) Usually vitamin derivatives
- d) Termed prosthetic group if bonded tight to their enzymes

e) All of the above are correct

:At V_{max} of an enzyme catalyzed reaction .8

- a) Further increase in substrate concentration increases the rate of the reaction
- b) All the enzyme molecules are combined with substrate
- c) The substrate concentration that produces maximal velocity is termed K_m
- d) Half the enzymes are combined with substrate
- e) K_m is maximum

In case of non-competitive inhibition of enzymes .9

- a) V_{max} can not be reached
- b) There is no significant alteration of the active site
- c) Inhibitor is structurally similar to the substrate
- d) Inhibitor can be removed by increasing substrate concentration
- e) V_{max} is increased

:Isoenzymes .10

- a) Are factors required for enzyme activity
- b) Are functional plasma enzymes
- c) Show similar responses to inhibitor molecules
- d) Are important tools in the diagnosis & prognosis of disease
- e) Catalyze oxidation-reduction reactions

:Hydrolases enzymes include .11

- a) Racemases and oxygenases
- b) Isomerases and lyases
- c) Peptidases and lipases
- d) Oxidoreductases and catalases
- e) Transaminases and hydrolases

?Which of the following statements is true of enzyme catalysts .12

- a) To be effective they must be present at the same concentration as their substrate
- b) They can increase the equilibrium constant for a given reaction by a thousand-fold or more
- c) They lower the activation energy for conversion of substrate to product
- d) Their catalytic activity is independent of pH
- e) They are generally equally active on D and L isomers of a given substrate

:Allosteric effector influences the enzyme activity by .13

- a) Covalently modifying the enzyme
- b) Binding to the substrate and altering its conformation
- c) Competing for catalytic site with substrate
- d) Changing the specificity of the enzyme for its substrate
- e) Binding to a site on the enzyme distinct from catalytic site

:The following enzyme complex requires both thiamine and lipoic acid .14

- a) Malate dehydrogenase
- b) Alpha-ketoglutarate dehydrogenase
- c) Fumarase
- d) Isocitrate dehydrogenase

e) Succinate dehydrogenase

:Stereoisomers .15

- a) Diversify physical and chemical properties of molecules
- b) Have the same spatial configuration
- c) Exhibit enzyme specificity
- d) Include epimers, anomers and enantiomers
- e) All A, C and D are correct

:In an enzyme catalyzed reaction .16

- a) Enzymes increase the energy barrier for the reaction
- b) Enzymes form a permanent complex with the substrate
- c) After the reaction, enzymes are recovered unchanged
- d) V-max is reached in the presence of a non-competitive inhibitor
- e) The equilibrium constant is decreased

:Isomerases .17

- a) Racemases and mutases
- b) Phosphatases and hydratases
- c) Lyases and oxidases
- d) Transaminases and kinases
- e) Hydratases and glucosidases

:Subclasses of oxidoreductases include .18

- a) Kinases and oxygenases
- b) Reductases and dehydrogenases
- c) Racemases and oxidases
- d) Catalases and aminotransferases
- e) Peroxidases and anomerase

:The active site of an enzyme .19

- a) Is rich with hydrophobic amino acid side chains
- b) Binds allosteric effectors
- c) Is similar in shape to the substrate
- d) Occupies most of the enzyme molecule
- e) None of the above

:Coenzymes .20

- a) Are small organic molecules needed for enzyme activity
- b) Are associated with apoproteins
- c) Are vitamin derivatives
- d) Are called prosthetic groups if they are tightly bound to the enzyme
- e) All of the above is correct

:In competitive inhibition of an enzyme .21

- a) K_m and V_{max} are decreased
- b) Inhibitor is structurally similar to the substrate
- c) Inhibition cannot be reversed by increasing substrate concentration
- d) V_{max} is decreased
- e) The inhibitor binds to the regulatory site on the enzyme

:In non-competitive inhibition of enzyme activity .22

- a) Both V_{max} and K_m are increased
- b) Inhibitor has no structural resemblance to the substrate
- c) There is a significant alteration of the active site
- d) V_{max} is decreased
- e) A is the only wrong answer

:The following statements describe Allosteric enzymes .23

- a) Effectors may enhance or inhibit substrate binding
- b) Binding of the substrate to the Allosteric site displaces the effectors
- c) They are always multimeric proteins
- d) The regulatory site is identical to the active site
- e) Binding of substrate to the active site releases allosteric effector

:The following statements about regulation of enzyme activity are correct .24

- a) Phosphorylation/ dephosphorylation is the common covalent modification
- b) Regulatory enzymes usually catalyze reversible reactions
- c) Allosteric inhibitors are usually the end products
- d) Increase in the enzyme level has no effect
- e) Irreversible inhibition is physiological type of regulation

:Phosphorylation/ dephosphorylation of enzymes .25

- a) Occurs on specific serine and threonine residues
- b) Is a covalent type of modification
- c) Is a reversible type of regulation
- d) Is achieved by the activities of protein kinases and phosphatases
- e) All of the above is correct

:The enzyme for starch digestion is .26

- a) Salivary amylase
- b) Pancreatic amylase
- c) Lysosomal glucosidase
- d) Both A and B
- e) All of the above is correct

:Hydrolases include .27

- a) Catalases and mutases
- b) Oxygenases and dehydrogenases
- c) Esterases and glucosidases
- d) Aminotransferases and racemases
- e) Epimerases and oxidases

:The following enzymes are absent in muscles but are present in the liver .28

- a) Pyruvate carboxylase and citrate synthase
- b) Glucose-6-phosphatase and glycerol kinase
- c) Lactate dehydrogenase and isocitrate dehydrogenase
- d) Pyruvate dehydrogenase and beta-ketothiolase
- e) Glycogen synthase and 3-HMG CoA synthase

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- d) Their catalytic activity is independent of pH
- e) They are generally equally active on D and L isomers of a given substrate

:Isoenzymes .30

- a) Are differentiated by dialysis
- b) Show the same electrophoretic mobility
- c) Include lactate dehydrogenase
- d) Are used as diagnostic tools
- e) Both C and D are correct

:The pancreatic enzyme not involved in protein digestion is .31

- a) Carboxypeptidase
- b) Trypsin
- c) Lipase
- d) Elastase
- e) Chymotrypsin

:Restriction enzymes .32

- a) Recognize specific nucleotide sequences in DNA
- b) Cut both strands of DNA
- c) Often produce single stranded tails
- d) Do all of the above
- e) Do none of the above

:Enzymes involved in hydrolysis .33

- a) Are called decarboxylases
- b) Use folic acid as a coenzyme
- c) Include dehydrogenases
- d) Include peptidases
- e) Always require coenzyme

:Enzymes and pH .34

- a) All enzymes have one optimum pH
- b) Enzymes in the stomach require alkaline pH
- c) All enzymes in eukaryotic cells require the same pH
- d) The optimum pH is the one at which the enzyme has maximum activity
- e) The pH is the same in the lysosome and plasma

:Isomerases .35

- a) Catalyze oxidation-reduction reactions
- b) Do not change the structural formula of the substrate
- c) Include transaminases
- d) Use water to break covalent bonds

e) Require ATP to form covalent bonds

:The following vitamin work as coenzymes with the corresponding enzyme .36

a) Biotin and carboxylases

b) Cobalamin with isomerases

c) Retinol with lyases

d) Riboflavin with hydrolases

e) Thiamin and isomerases